

Ani Abrahamyan

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EDUCATION

Purdue University

Aug 2025 – May 2029

B.S. Computer Engineering, Minors in Math & Philosophy | GPA: 3.54/4.0

West Lafayette, IN

- **Relevant Coursework:** Advanced C Programming, STARS Chip Design Skills, Semiconductor Seminar, Multivariable Calculus
- **Awards and Honors:** Dean's List, Gilbert J. Easley Scholarship in ECE, InnovateHer Finance Track 1st Place
- **Activities:** Executive Board of Armenian Student Association, Delta Phi Lambda Sorority

EXPERIENCE

STARS Chip Design Skills Team Member

June 2026 – Aug 2026

Purdue System-on-Chip Extension Technologies

Remote

- Designed finite state machines in SystemVerilog for DMA-style memory-to-memory transfer control, building self-checking testbenches for verification
- Designed and verified a memory-mapped popcount hardware accelerator in SystemVerilog, achieving a 4.9x speedup over software execution
- Debugged and validated RTL designs in Verilator and GTKWave, tracing waveforms to confirm FSM behavior against expected outputs
- Explored RTL-to-gate-level synthesis, analyzing netlists and area reports to evaluate tradeoffs between sequential and combinational logic

Solution Engineer Intern

June 2026 – Present

Cummins Inc.

Columbus, IN

- Architecting a two-agent AI system in Databricks to automate data harmonization and undefined-value resolution across 5+ Cummins ERP instances, replacing manual data steward review with a confidence-scored pipeline
- Built a training/testing framework validating AI agent output against expected ground-truth values, tracking accuracy distributions and confidence scores prior to production scale-out
- Coordinated with data lake administration and business stakeholders to secure production data access and align on schema design, unblocking agent development timelines

Software Engineer Intern

Dec 2025 – Jan 2026

Canary Labs

Martinsburg, PA

- Applied Agile development methodologies and DevOps workflows to develop C# applications while managing version control and deployment throughout the software development lifecycle
- Tracked data flow from field collectors to remote historians, supporting large-scale industrial analytics

PROJECTS

Single-Cycle RISC-V CPU | *SystemVerilog, RISC-V*

[GitHub](#)

- Developing a single-cycle RISC-V processor core in SystemVerilog targeting the RV32I instruction set, including the datapath, register file, ALU, immediate generator, and control decoders for arithmetic, branch, jump, and memory instructions
- Working toward instruction-level verification with self-checking testbenches and assembly test programs in Verilator, using waveform debugging to validate decode and control-logic behavior

Systolic Array for 2D Matrix Multiplication | *SystemVerilog, Verilator, C++*

[GitHub](#)

- Architecting an output-stationary systolic array in SystemVerilog for $N \times N$ matrix multiplication, with cycle-accurate operand synchronization across an N^2 MAC grid
- Designing shift-register-based skew logic to align operand timing across the processing element grid
- Verifying functional correctness via self-checking Verilator/C++ testbench, comparing against a reference model

TECHNICAL SKILLS

Languages: C, C++, Python, SystemVerilog, MATLAB

Hardware Design: RTL Design, Digital Logic, RISC-V, FSM Design

Verification: Self-Checking Testbenches, Verilator, GTKWave, Waveform Debugging

Tools: Git, Linux (CLI), LTSpice, Databricks, Power BI